

Partial Least Square

- ↳ It is statistical and machine learning method.
- ↳ It find linear relationship b/w independent variable and dependent p. variable.
- ↳ by projecting them into new latent spaces that maximize covariance.
- ↳ It works best when many highly correlated predictor variable.
- ↳ It designed to cope with problems in data specifically, small datasets, missing values and multicollinearity.
- ↳ ~~It is~~ PLS is a regression method allows for the identification of underlying factor, which is linear combination of explanatory variable or latent variable.
- ↳ It is considered to be a better alternative to multiple linear regression and PCA regression since it provides for more robust model parameters that do not change with new calibration sample from population.
- ↳ PLS ~~is~~ is an improvement on PCA.

↳ PLS is supervised multivariate regression, dimensionality reduction technique.

↳ It projects both predictor variable (X) and response variable (Y) into a lower-dimensional latent space to find components that maximize the covariance b/w them.

↳ PLS is specially specifically engineered for situation where predictors are highly collinear or vastly outnumber observation ($P \gg n$)

↳ Shrinks a large set of explanatory variable (X) into a smaller set of uncorrelated latent components.

↳ ~~Maximizes~~ Unlike PCA, which only maximizes variance in X , PLS maximizes the covariance b/w the components of X and the response ~~response~~ variable Y .

↳ The drawback of PCR (principal component Regression) is that it weights predictor variable (X) according to variance, and not correlation with target variable (Y). PLS overcome these drawback.

↳ PLS look for factor showing good covariance with Y . This favors both accuracy and robustness.

PLS work actually as follow.

The important idea is that PLS did not directly use x_1 and x_2 separately

$$t = 0.4472x_1 + 0.8944x_2$$

$$y = 1.3416t + 6$$

$$y = 1.3416(0.4472 + 0.8944x_2) + 6$$

$$y = 0.6x_1 + 1.2x_2 + 6$$

PLS work flow

⊗

$$X, Y \rightarrow \text{centre/scale} \rightarrow w \rightarrow t = Xw \rightarrow q \rightarrow \hat{y} = tq + \bar{y}$$

The General Underlying model of multi-variate PLS with L components is

$$X = TP^T + E$$

$$Y = UQ^T + F$$

where, X is an $n \times m$ matrix of predictors

• Y " " $n \times p$ " " response

• $T \in U$ are $n \times l$ " that respectively
Projection of X (X score, component or
factor matrix) and
Projection of Y (the Y score)

• $P \in \Theta$ are respectively $m \times l$ and $p \times l$ loading
matrices; • $E \in F$ are error term.

Diff. b/w PCA & PLS

Feature	PCA	PLS
Type	Unsupervised	Supervised
Goal	Maximize variance of features	Maximize correlation with target
Work well for	Clustering, data visualization	Regression, prediction
Relationship with target	Ignores it	Uses it to find relevant components.

PLS work

- ↳ PLS find new components (PLS1, PLS2...) that balance feature variance and correlation with the target variable.
- ↳ Standardize the data: convert all features to a similar scale.
- ↳ Find latent variables (PLS components): Identify new feature combinations that:
 - Capture variance (like PCA)
 - Are highly correlated with the target variable.
- ↳ Transform data into new components - Instead of using raw features, we use these new PLS components.
- ↳ Build a regression model using these transformed

Components.

Uses of PLS in ML

- ↳ Predicting chemical properties ex medicine effectiveness
- ↳ Financial forecasting (ex predicting stock price)
- ↳ Biology & genetics
- ↳ Industrial application

PLS mathematical representation.

Step 1: Standardized X and Y

$$x_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}, \quad y_i = \frac{y_i - \bar{y}}{s_y}$$

Step 2: Find the weight

$$w_1 \propto X^T Y, \quad w_1 = \frac{X^T Y}{\|X^T Y\|} \quad (\text{Normalized})$$

Step 3: Calculate score

$$t_1 = X w_1$$

Step 4: Calculate X -loading

$$p_1 = \frac{X^T t_1}{t_1^T t_1}$$

Step 5: Calculate Y -loading

$$q_1 = \frac{Y^T t_1}{t_1^T t_1}$$

Step 6: Deflate X

$$X \leftarrow X - t_1 p_1^T$$

Step 7: Deflate Y

$$Y = Y - t_1 q_1$$

Step 8: Repeat the ~~for~~ process for
 $t_2, t_3, \dots, t_k$

Step 9: PLS regression equation

After extracting k components, the final regression model can be written as:

$$\hat{Y} = T Q^T$$

Since

$$T = X W^*$$

the final PLS regression can be expressed as:

$$\hat{Y} = X B$$

where B is the PLS regression coefficient matrix

For a single response variable

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

The important point is that the coefficients are obtained through PLS latent-component procedure rather than directly estimation

ex

x_1	x_2	y
1	2	3
2	4	6
3	6	9

Clearly $y = x_1 + x_2$ approximately, how PLS obtain a latent component

Steps:

$$\bar{x}_1 = \frac{1+2+3}{3} = 2$$

$$\bar{x}_2 = \frac{2+4+6}{3} = 4$$

$$\bar{y} = \frac{3+6+9}{3} = 6$$

$$x_1 = x_1 - \bar{x}_1 \quad x_2 = x_2 - \bar{x}_2 \quad y = y - \bar{y}$$

$$\begin{matrix} -1 \\ 0 \\ 1 \end{matrix}$$

$$\begin{matrix} -2 \\ 0 \\ 2 \end{matrix}$$

$$\begin{matrix} -3 \\ 0 \\ 3 \end{matrix}$$

$$X = \begin{bmatrix} -1 & -2 \\ 0 & 0 \\ 1 & 2 \end{bmatrix} \quad Y = \begin{bmatrix} -3 \\ 0 \\ 3 \end{bmatrix}$$

Step 2: Calculate the PLS weight vector

$$w = X^T Y$$

$$X^T = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \end{bmatrix}$$

$$w = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \end{bmatrix} \times \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \end{bmatrix} \quad \|w\| = \sqrt{6^2 + 12^2} = 13.416$$

$$\text{Hence } w = \begin{bmatrix} \frac{6}{13.416} \\ \frac{12}{13.416} \end{bmatrix} = \begin{bmatrix} 0.4472 \\ 0.8944 \end{bmatrix}$$

Step 3

Calculate the latent variable t

$$t = Xw$$

$$t = \begin{bmatrix} -1 & -2 \\ 0 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 0.4472 \\ 0.8944 \end{bmatrix}$$

$$t_1 = (-1) * 0.4472 + (-2) * 0.8944 \\ = -2.236$$

$$t_2 = 0$$

$$t_3 = (1) (0.4472) + (2) (0.8944) \\ = 2.236$$

$$t = \begin{bmatrix} -2.236 \\ 0 \\ 2.236 \end{bmatrix}$$

$$Y = tq^T + E$$

Step 4

$$q = \frac{t^T Y}{t^T t}$$

$$t^T Y = (-2.236)(3) + (0)(0) + (2.236)(3) \\ = 13.416$$

$$t^T t = (-2.236)^2 + 0^2 + (2.236)^2 \\ = 10$$

$$q = \frac{13.416}{10} = 1.3416$$

Step 5 predict $\hat{y}$

$$\hat{y} = xq$$

$$= \begin{bmatrix} -2.236 \\ 0 \\ 2.236 \end{bmatrix} (1.3416)$$

$$= \begin{bmatrix} -3 \\ 0 \\ 3 \end{bmatrix}$$

~~$\hat{y}$~~
we add original mean

$$\bar{y} = 6(2) + 5(5) + 0(1) = 37$$

Hence the original scale prediction are

$$y_{\text{original}} = \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$$

Thus PLS predicts the target perfectly for this simple example.